

EDUCATION

DePaul University

Masters of Science in Computer Science (AI), GPA: 3.80

Chicago, United States

Sept 2024 - July 2026

Kathmandu University

Bachelor of Science in Computer Science

Dhulikhel, Nepal

Feb 2016 - June 2020

SKILLS

Languages: Python, SQL, R

ML/AI: TensorFlow, scikit-learn, PyTorch, Pandas, NumPy, CNNs, Transformers

Cloud & Big Data: AWS (EMR, S3, SageMaker), Apache Spark, Hadoop

Tools & Specializations: Git, Docker, Jupyter, Oracle Data Integrator (ODI), ETL Pipelines, Computer Vision, NLP, MLOps

PROFESSIONAL EXPERIENCE

DePaul University

AI Research Assistant

Chicago, United States

July 2025 - Present

- Researching scientific chart mining with DB/DB++/PSENet/EAST detection and SVTR recognition using PPHGNetV2_B4, achieving 97% accuracy through preprocessing optimization and dataset fine-tuning.

Cotiviti Nepal Pvt. Ltd.

Software Engineer

Kathmandu, Nepal

July 2021 - July 2023

- Built ETL pipelines using SQL, PL/SQL stored procedures, and ODI to process US healthcare PHI data for 15+ clients, designing automated validation checks to ensure data quality throughout client onboarding.
- Collaborated with cross-functional teams (developers, QA, PMs) to deploy production solutions and mentored 10+ team members on ODI workflows.

Kathmandu University

Graduate Research Assistant

Dhulikhel, Nepal

Mar 2021 - Apr 2022

- Developed Nepali image captioning model using CNN and Transformer architectures, optimizing performance through hyperparameter tuning and attention mechanisms using BLEU and METEOR metrics.

PROJECTS

Delivery Route Optimization, *AI search algorithms for logistics*

July 2025 - Aug 2025

- Compared BFS, UCS, and A* algorithms on real-world delivery scenarios, achieving 4x performance improvement with A* through custom heuristics while guaranteeing optimal solutions.

Developer Programming Language Prediction, *Production-ready ML pipeline*

June 2025 - July 2025

- Achieved 77.5% test accuracy in a production-ready ML pipeline for developer programming language classification, using grid search + k-fold CV across Decision Tree, Random Forest & Logistic Regression.

Applied ML Implementations, *25+ Applied ML Projects*

Nov 2024 - May 2025

- Implemented diverse ML solutions including fraud detection with imbalanced data (98% recall), cancer classification (96% F1-score), customer segmentation, geospatial clustering, and ensemble methods with comprehensive evaluation frameworks.

Evolution Simulator, *A demonstration of genetic algorithm*

Nov 2024 - May 2025

- Built a simulation where bipedal creatures evolve to walk using genetic algorithms. Featured on aijs.rocks.

ACHIEVEMENTS

- Published Research Paper: "Evaluating the Application of SOLID Principles in Modern AI Framework Architectures" on arXiv (2025)
- Associate Member, Sigma Xi, The Scientific Research Honor Society (2025)